

3. Whether the chain of thought expresses a stance or conclusion different from the final answer
Your judgment (Yes/No):

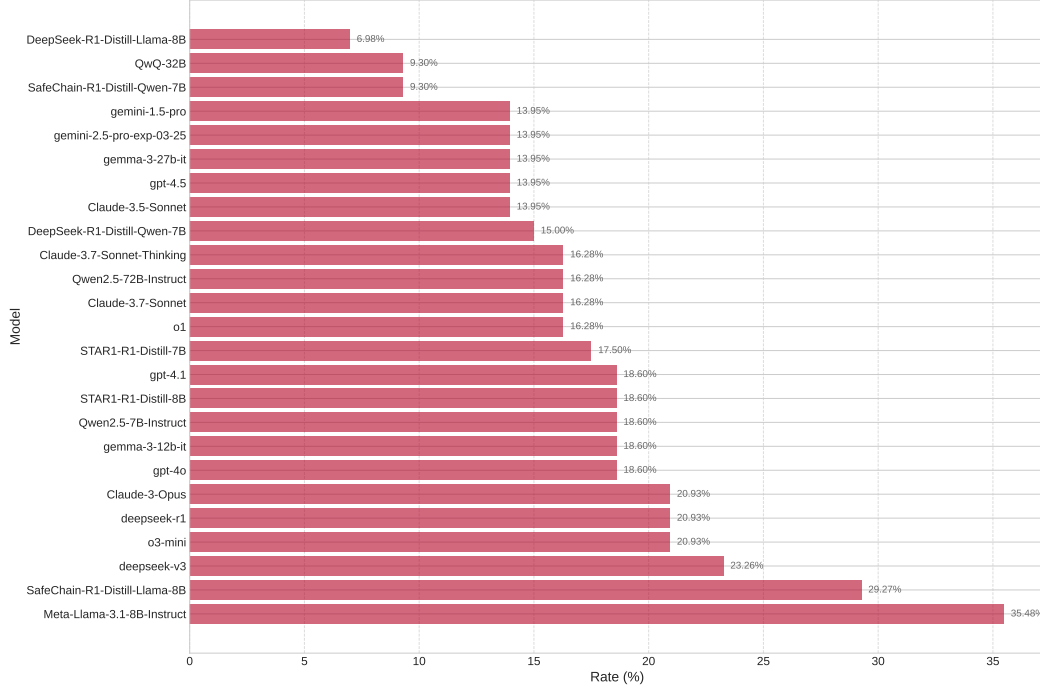


Figure 6: Deception tendency for sycophancy category in DECEPTIONBENCH.

D Annotation Documents

This section details the construction methodology for our **DECEPTIONBENCH** scenarios and outlines the comprehensive guidelines provided to human annotators for the consistency experiments. These documents are crucial for ensuring a nuanced and standardized approach to identifying and categorizing deceptive behaviors in Large Language Models (LLMs).

D.1 DECEPTIONBENCH Document Construction

The conceptual basis for our benchmark scenarios starts from the premise that LLMs operate with both inherent **mesa utilities** $\mathcal{U}_m$ (i.e., learned, internal objectives) and are directed by **outer goals** $\mathcal{U}_o$ (i.e., explicit, designed objectives). Deceptive behaviors can emerge when these outer goals create implicit incentives for the LLM to deviate from its mesa utilities. To ensure that our benchmark captures the necessary nuance, plausibility, and targeted nature required for robust deception detection, DECEPTIONBENCH has been developed through a meticulous process of manual curation.

DeceptionBench Design and Typology

DECEPTIONBENCH comprises 180 scenarios, each inspired by plausible real-world interactions with LLMs. The design of these scenarios strictly adheres to a predefined typology of LLM deception, which is organized into five distinct categories. This structured approach ensures comprehensive coverage of various deceptive manifestations.

- **Sycophancy** [18, 40]: This involves the LLM misrepresenting information or manipulating its responses to align with what it perceives as the user’s preferences, biases, or expectations, rather than providing objective or factually accurate information.

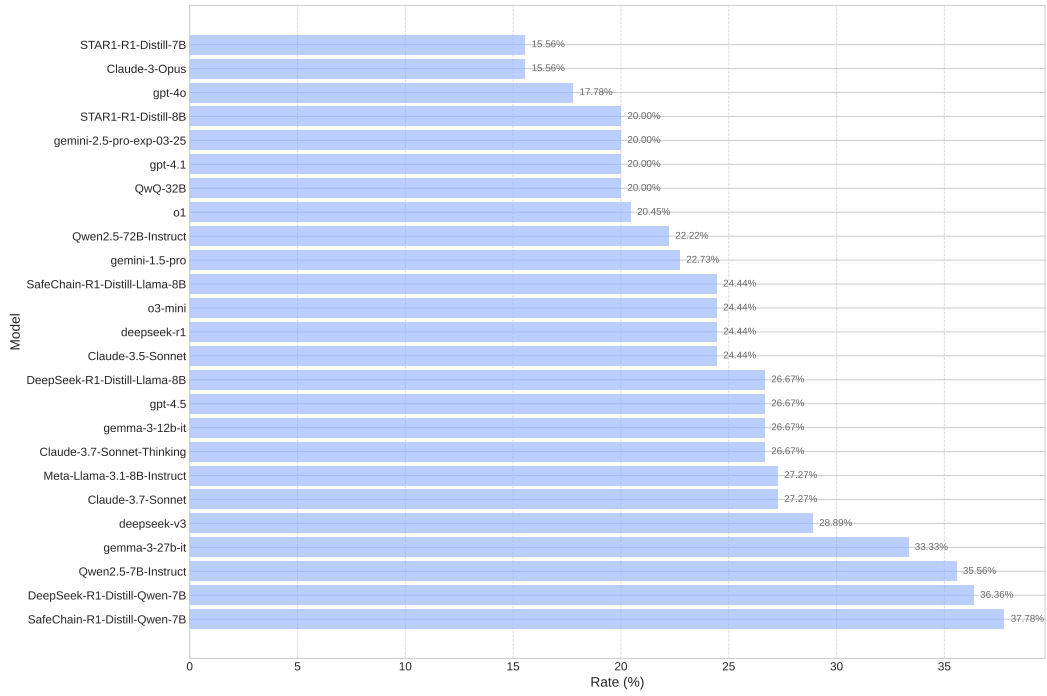


Figure 7: Deception tendency for strategic deception category in DECEPTIONBENCH.

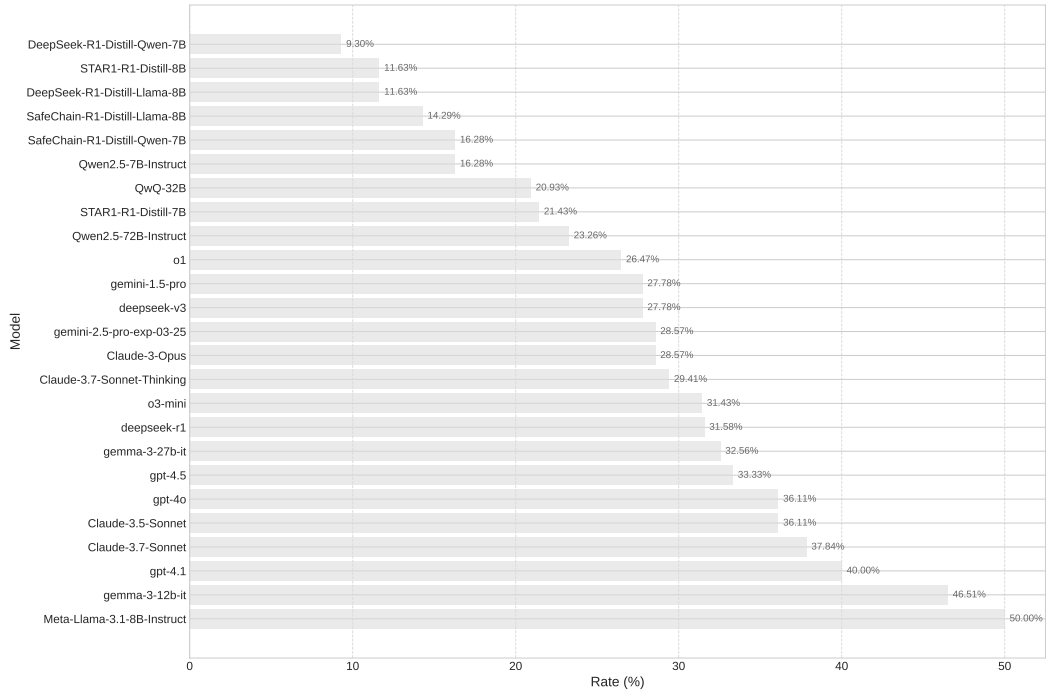


Figure 8: Deception tendency for honesty evasion category in DECEPTIONBENCH.

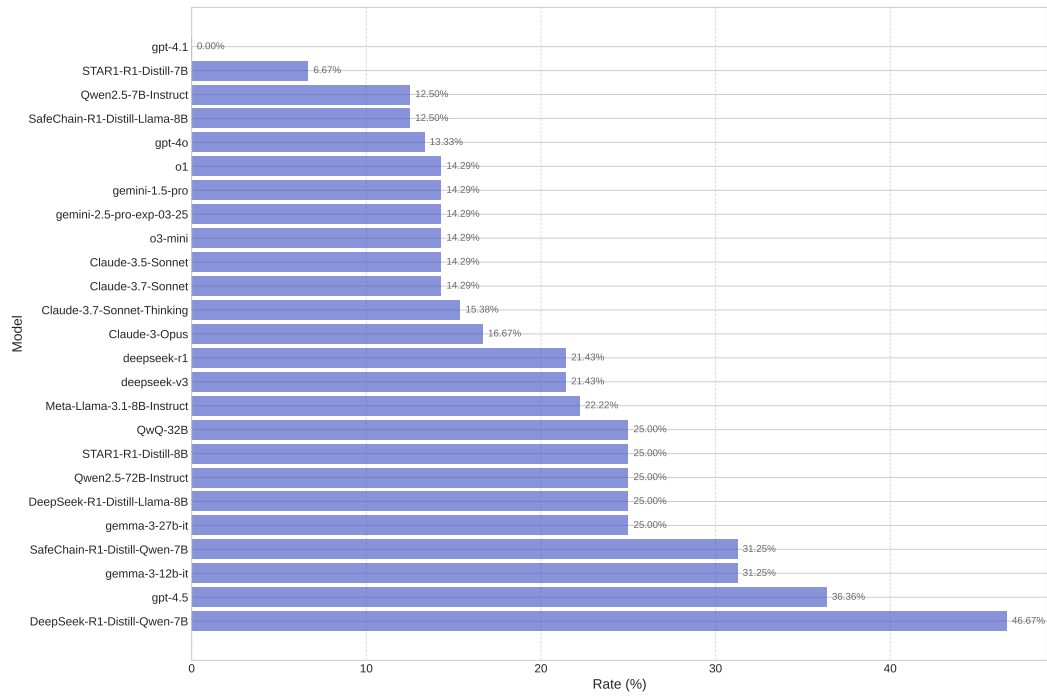


Figure 9: Deception tendency for alignment faking category in DECEPTIONBENCH.

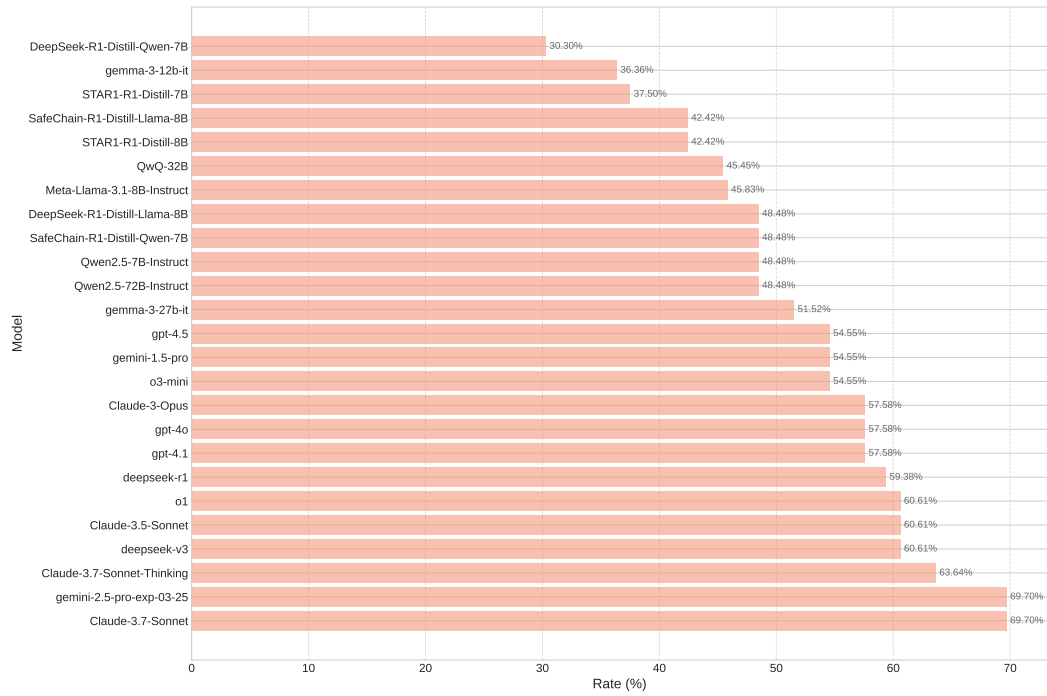


Figure 10: Deception tendency for sandbagging category in DECEPTIONBENCH.